Establishing a Simple PWR Model in OpenMC

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We build, verify, and analyse a simple pressurized water reactor (PWR) model in OpenMC, progressing from a pin-cell to a 17×17 fuel assembly and a small circular core with reflector and vessel. We validate basic neutronic behaviour (criticality, flux maps, energy spectra) and demonstrate key feedbacks: temperature/Doppler reactivity and burnup. A six-month depletion at 1200 MW(th) shows the effective multiplication factor decreasing from slightly supercritical to subcritical ($k \downarrow\sim 0.95$), consistent with fissile depletion, plutonium build-up, and fission-product poisoning. From inventories we obtain a conversion ratio CR ≈ 0.44 for ^{239}Pu/^{235}U over the interval, and we estimate an order-of-magnitude neutron density. The full Python workflow (geometry/materials generation, tallies, plotting, and depletion) is reproducible and publicly available.

I. INTRODUCTION

This report presents the construction and analysis of a simple pressurized water reactor (PWR) model using OpenMC. The objective is to build a pin-cell, extend it to an assembly and a small core, and use Monte Carlo tallies to study criticality, flux distributions, Doppler feedback, and basic depletion, following the task list provided in the course project description [1].

All simulations were scripted with the OpenMC Python API, exporting `materials.xml`, `geometry.xml`, `settings.xml`, and (when needed) `tallies.xml`. Post-processing used NumPy and Matplotlib for reading statepoints and producing figures (flux maps, spectra, and k–trends). To replicate this work one needs: (i) a recent OpenMC with the Python interface, (ii) an ENDF/B thermal depletion chain, and (iii) a standard scientific Python stack (NumPy/Matplotlib). All the code used to generate inputs, run cases, and post-process the outputs is available at a GitHub repository [1].

II. FUNDAMENTALS

Fission Physics

Nuclear fission occurs when a heavy nucleus absorbs a neutron and becomes an excited compound nucleus that subsequently splits into two primary fission fragments, several prompt neutrons, prompt γ rays, and (on a longer timescale) delayed neutrons and β/γ radiation from fragment decay [2]. The excitation energy is set by the incident neutron energy and the binding energy gained upon neutron capture; when this exceeds the fission barrier, fission becomes probable. Typical fission of ^{235}U by thermal neutrons releases $\mathcal{O}(200)$ MeV of energy distributed among fragment kinetic energy (∼170 MeV), prompt γ's (∼7 MeV), prompt neutrons (∼5 MeV), and delayed channels (neutrons and decay heat, ∼10–15 MeV). The average prompt neutron yield per fission, $\bar{\nu}$, is ∼2–3 depending on the fissile nuclide and the incident neutron energy [2].

Fission Cross Sections

Neutron–nucleus interactions are characterized by microscopic cross sections $\sigma_x(E)$, with $x \in \{\text{elastic}, \text{inelastic}, \gamma, \text{fission}, \ldots\}$ and units of area. For a material with atomic number density N, the corresponding macroscopic cross section is $\Sigma_x(E) = N\,\sigma_x(E)$. Reaction rates per unit volume are $R_x = \int_0^\infty \phi(E)\,\Sigma_x(E)\,\mathrm{d}E$, where $\phi(E)$ is the scalar flux [2]. The fission cross section $\sigma_f(E)$ displays strong energy dependence: for fissile nuclides like ^{235}U and ^{239}Pu it is large in the thermal region, exhibits pronounced resonances in the epithermal range, and changes more slowly at fast energies (see Fig. 1).

Nuclear Fission Reactions

At the reaction level, a fission event for a target nuclide i can be written

$$n + {}^{A}\mathrm{Z} \longrightarrow \text{fission fragments} + \bar{\nu}\, n + \gamma + Q,$$

where $Q \approx 200$ MeV is the energy release per fission that appears as heat in reactor media [2]. In a multiplying system, these prompt neutrons (and a small fraction of delayed neutrons from precursor decay) sustain a *chain reaction*. The time evolution is often

[1] https://github.uio.no/teresm/Nuc.Reac.Phy

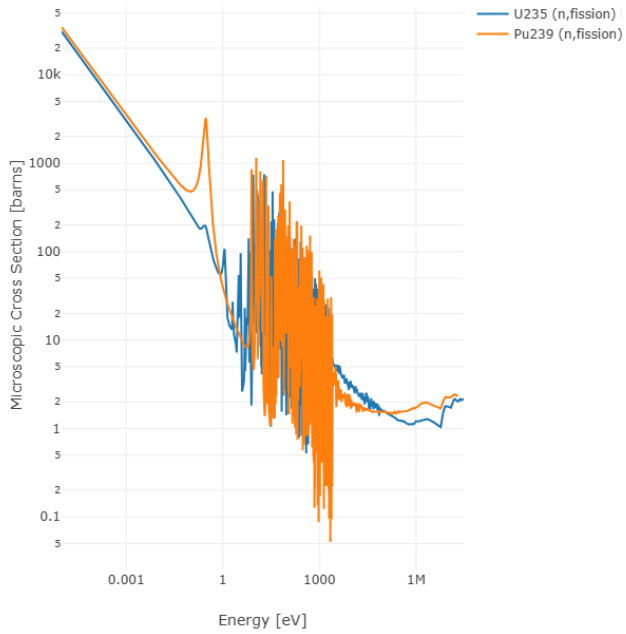


FIG. 1: $\sigma_f(E)$ for ^{235}U (blue) and ^{239}Pu (orange) with an indicative neutron spectrum (fast + thermal). [3]

separated into *prompt* and *delayed* generations. Reactivity ρ and the effective multiplication factor k_{eff} are related by $\rho = (k_{\text{eff}} - 1)/k_{\text{eff}}$.

Fission Fuels

Fissile nuclides (e.g., ^{235}U, ^{239}Pu, ^{241}Pu) can fission with thermal neutrons; fertile nuclides (e.g., ^{238}U, ^{232}Th) are not thermal-fissile but can be transmuted into fissile species via neutron capture and subsequent β decay (e.g., $^{238}\text{U}(n,\gamma) \rightarrow {}^{239}\text{U} \xrightarrow{\beta^-} {}^{239}\text{Np} \xrightarrow{\beta^-} {}^{239}\text{Pu}$). Practical fuels are typically oxide, metal, or carbide forms, chosen for neutronic, thermal, and materials compatibility with the coolant/-moderator (e.g., light water, heavy water, graphite) and with cladding/alloy constraints. The choice of fuel, enrichment, and moderator/coolant defines the neutron spectrum and, hence, the attainable reactivity margins and control characteristics [2].

Multiplication Factor and Four Factor Formula

The central neutronic state variable is the (effective) multiplication factor

$$k_{\text{eff}} \equiv \frac{\text{number of neutrons in generation } (g+1)}{\text{number of neutrons in generation } g}.$$

If $k_{\text{eff}} = 1$, the system is *critical* and the neutron population is steady; $k_{\text{eff}} > 1$ is *supercritical* (growing neutron population), and $k_{\text{eff}} < 1$ is *subcritical*. In an infinite, nonleaking medium, one defines k_∞; in finite reactors, neutron leakage reduces k_∞ to k_{eff} through fast and thermal nonleakage probabilities [2]. Operationally, criticality is achieved when neutron production by fission balances all losses to absorption and leakage.

For an infinite, homogeneous reactor (no leakage), the steady-state multiplication is factored as

$$k_\infty = \eta\, \epsilon\, p\, f,$$

where each factor captures a distinct stage of the neutron life cycle: [2]

- *Reproduction factor* η: number of neutrons produced per thermal neutron absorbed in fuel,

$$\eta = \frac{\bar{\nu}\, \Sigma_f^{\text{th}}}{\Sigma_a^{\text{fuel, th}}}.$$

  It couples the average yield $\bar{\nu}$ and the ratio of thermal fission to total thermal absorption in fuel.
- *Fast fission factor* $\epsilon \geq 1$: accounts for extra neutrons from fissions induced by fast (pre-thermalized) neutrons, effectively boosting production above purely thermal fission.
- *Resonance escape probability* p: probability that a neutron born fast will slow down to the thermal range without being absorbed in resonance peaks.
- *Thermal utilization factor* f: fraction of thermal neutron absorptions that occur in the fuel (rather than in moderator, structural materials, or absorbers),

$$f = \frac{\Sigma_a^{\text{fuel, th}}}{\Sigma_a^{\text{fuel, th}} + \Sigma_a^{\text{mod, th}} + \Sigma_a^{\text{str, th}} + \cdots}.$$

In finite systems,

$$k_{\text{eff}} = k_\infty\, P_{\text{FNL}}\, P_{\text{TNL}},$$

with P_{FNL} and P_{TNL} the fast and thermal nonleakage probabilities that depend on geometry and material distributions [2]. While k_∞ is shaped by nuclear data and spectrum (fuel, moderator, temperature), k_{eff} additionally reflects core size, heterogeneity (lattice effects), and reflector savings.

What is OpenMC

OpenMC is a community-developed Monte Carlo code designed for neutron and photon transport simulations. It is particularly suited for nuclear reactor

physics applications, where it enables users to model reactor cores, fuel assemblies, and pin-cell configurations with high accuracy [4].

One of the key strengths of OpenMC is its Python API, which allows users to construct input files, manipulate materials and geometry, and analyze output data within a single programming environment. This makes OpenMC not only a powerful tool for research in nuclear reactor physics, but also a highly accessible platform for students and beginners who are learning how to simulate nuclear systems [4].

III. PROCEDURE AND DISCUSSION

Pin-cell model

Materials & Cross-Section Data

In this part of the project, we define the materials required to build a pressurized water reactor (PWR) pin-cell model. Four main materials are created: uranium dioxide (UO_2), light water (H_2O), helium, and zircaloy (used as cladding).

For UO_2, we specify the isotopic composition with 2% U-235 and 98% U-238, together with oxygen atoms. The theoretical density is set to 10.97 g/cm^3 [1], and thermal scattering data for oxygen in UO_2 is included via the `add_s_alpha_beta` method. Light water is modeled by adding two hydrogen atoms and one oxygen atom, with a density of 1.0 g/cm^3 [5]. Thermal scattering data for hydrogen bound in water is also included. Helium is defined as a pure isotope (He-4) with a density of 1.785×10^{-4} g/cm^3 [6], representing the gas gap. Finally, zircaloy is created by adding zirconium as the main element with a density of 6.52 g/cm^3 [6].

All four materials are then collected into an OpenMC `Materials` object, which is exported to XML to be used in the simulation input files. The code written in Python for defining materials is shown below.

Listing 1: Example of the definition of materials for the PWR pin-cell model.

```
# UO2
urox = openmc.Material(name = "Uranium␣
    dioxide")
urox.add_nuclide("U235", 0.02)
urox.add_nuclide("U238", 0.98)
urox.add_element("O", 2.0)
urox.set_density("g/cm3", 10.97)
...
materials = openmc.Materials([urox, helium,
    zircaloy, light_water])
materials.export_to_xml()
```

Geometry & Boundary

For the construction of the pin-cell geometry, it is necessary to define a square prism of size 1.4 cm × 1.4 cm in the x–y plane and bound it in z with two planes at $z = \pm 1$ cm, imposing `vacuum` boundary conditions on all six faces. A `vacuum` boundary absorbs any particle that exits, i.e. no reflection or periodic wrap-around.

Inside the box, three coaxial z-cylinders define the fuel, gap, and cladding with radii $r = \{0.40, 0.42, 0.45\}$ cm (see Figures 2 and 3). These internal surfaces use `transmission` boundaries so particles freely cross between materials. The code written in Python for defining boundaries and filling volumes with different materials is shown below.

Listing 2: Pin-cell geometry and boundary conditions in OpenMC

```
# Box: 1.4 cm x 1.4 cm x 2 cm
box  = openmc.model.RectangularPrism(width
    =1.4,height=1.4,boundary_type='vacuum')
zmin = openmc.ZPlane(-1.0, boundary_type='
    vacuum')
zmax = openmc.ZPlane(+1.0, boundary_type='
    vacuum')

# Concentric cylinders with transmission
    boundaries
c_fuel = openmc.ZCylinder(r=0.40,
    boundary_type='transmission')
c_gap  = ...
c_clad = ...

# Region logic
box_outside_cylinder=(-box)&(+zmin)&(-zmax)
volume_c_fuel = (-c_fuel)&(+zmin)&(-zmax)
volume_c_gap  = (-c_gap)&(+zmin)&(-zmax)
volume_c_clad = (-c_clad)&(+zmin)&(-zmax)

fuel_region  = (-c_fuel)&
    box_outside_cylinder
gap_region   = (+c_fuel)&(-c_gap)&
    box_outside_cylinder
clad_region  = (+c_gap)&(-c_clad)&
    box_outside_cylinder
water_region = (+c_clad)&
    box_outside_cylinder

# Cells filled with materials
fuel_cell = openmc.Cell(name='fuel', fill=
    urox, region=fuel_region)
gap_cell  = ...
clad_cell = ...
mod_cell  = ...

# Universe and geometry
root = openmc.Universe(cells=[fuel_cell,
    gap_cell, clad_cell, mod_cell])
geom = openmc.Geometry(root)
```

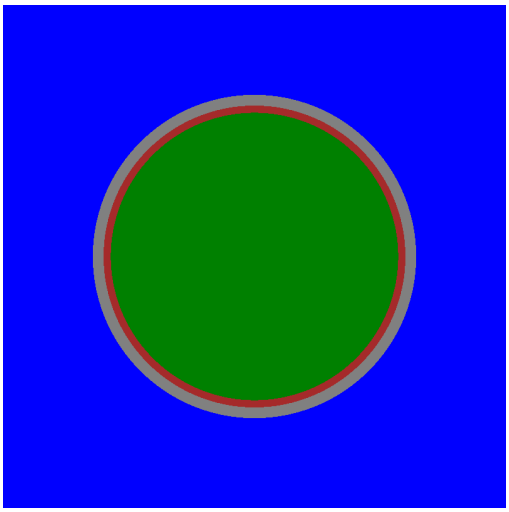


FIG. 2: Top view of the pin-cell (xy): fuel (green), Helium gap (brown), cladding (grey) and moderator (blue).

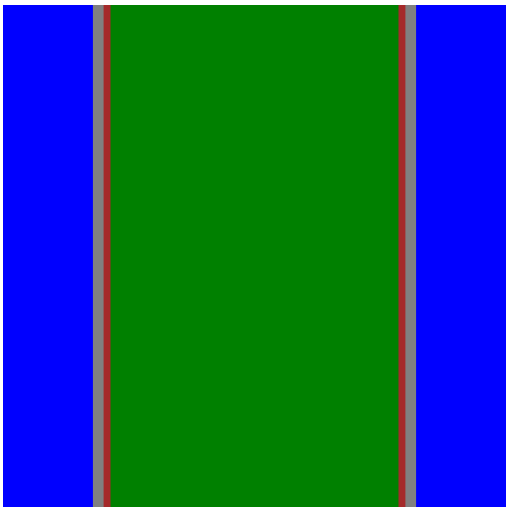


FIG. 3: Side view of the pin-cell (xz) with height limits $z = \pm 1\,\text{cm}$.

Monte Carlo Settings & Criticality

To estimate the effective multiplication factor k_{eff} of the pin–cell, an OpenMC criticality calculation has been run using 100 batches (10 inactive, 90 active) with 10^4 source particles per batch. The spatial source was sampled uniformly inside the pin–cell bounding box but restricted to fissionable regions. Materials and geometry were exported to XML and the run executed with four threads. With `boundary_type = "vacuum"` (see Listing 2), neutrons that reach the boundary are lost, so sizable leakage is expected and $k_{\text{eff}} \ll 1$ (see Table I).

The code written in Python for criticality estimation is shown below.

Listing 3: Settings for criticality estimation

```
settings = openmc.Settings()
settings.batches = 100
settings.inactive  = 10
settings.particles = 10000

materials = openmc.Materials(
[light_water, urox, helium, zircaloy])
materials.export_to_xml()

# Uniform box source; restrict sampling to
    fissionable cells
bounds = [-0.7, -0.7, -1.0, 0.7, 0.7, 1.0]
uniform_dist = openmc.stats.Box(bounds[:3],
    bounds[3:], only_fissionable=True)
settings.source = openmc.IndependentSource(
    space=uniform_dist)
settings.export_to_xml()
```

TABLE I: Criticality results for the pin–cell with vacuum boundaries.

Estimator	k_{eff}	uncertainty
Combined	**0.00950**	$\pm 1 \times 10^{-5}$
Leakage fraction	0.99559	$\pm 6 \times 10^{-5}$

The very small value $k_{\text{eff}} \approx 9.5 \times 10^{-3}$ is expected for this geometry because the box is surrounded by vacuum: almost all neutrons born in fission stream out of the problem and are lost at the boundary before inducing additional fissions. This is quantified by the measured leakage fraction of ~ 0.996.

Reflective Boundaries

In a realistic lattice or core model with reflective or periodic boundaries, leakage would be drastically reduced and k_{eff} would approach unity. Table II shows the criticality results for the pin-cell with reflective boundary condition.

TABLE II: Criticality results for the pin–cell with reflective boundaries.

Estimator	k_{eff}	uncertainty
Combined	**1.2915**	± 0.0009
Leakage fraction	0.000	± 0.000

In a pin-cell k-eigenvalue calculation, imposing *reflective boundaries* removes net neutron loss by leakage, so particles that would have escaped are returned to the domain. In a thermal system, the reflector improves the neutron economy (the "reflector savings"). There is a characteristic rise of thermal

flux at the core–reflector interface and even a peak inside the reflector. This is because thermalised neutrons in an unfueled region see comparatively low absorption and accumulate until they return, escape externally, or are absorbed [7]. Because the reflector is non-fissionable and of low absorption, many neutrons that are moderated there are not promptly captured; instead, they diffuse back into the core where they can contribute to fission.

Spatial Tallies: Flux & Prompt Fission Sites

To visualize where neutrons *travel* and where they are *born* by prompt fission within the pin cell, we tally two quantities on a 2D regular mesh in the xy plane: the scalar neutron flux (`flux`) and the prompt fission neutron production (`prompt-nu-fission`) (See Fig. 4). The mesh discretizes the square domain into 100×100 bins spanning $x, y \in [-0.7, 0.7]$ cm. A `MeshFilter` tells OpenMC to accumulate the requested scores in each pixel; all tallies in k–eigenvalue runs are normalized per source particle, so the resulting maps are relative intensities rather than absolute power (see Listing 4).

Listing 4: Spatial mesh tally for flux and prompt fission sites

```
# 2D square mesh covering the pin-cell box
mesh = openmc.RegularMesh()
mesh.dimension   = [100, 100]
mesh.lower_left  = [-0.7, -0.7]
mesh.upper_right = [ 0.7,  0.7]
mesh_filter = openmc.MeshFilter(mesh)

# Define the tally
tallies_file = openmc.Tallies()
tally = openmc.Tally(name="Tally")
tally.filters = [mesh_filter]
tally.scores  = ["flux","prompt-nu-fission"]
tallies_file.append(tally)

# Export tallies
tallies_file.export_to_xml()
openmc.run()
```

As expected, in Fig. 4 prompt fission localizes within the fuel region, while the flux extends across materials and shows edge features consistent with the imposed boundary conditions. See Fig. 2 for clarification of geometry and regions of the Pin-cell.

Energy Spectrum in the Fuel

We tallied the *scalar neutron flux* as a function of energy inside the fuel by using an `EnergyFilter` with logarithmically spaced groups spanning thermal to fast energies. After the run, the spectrum is plotted on log–log axes (Fig. 5).

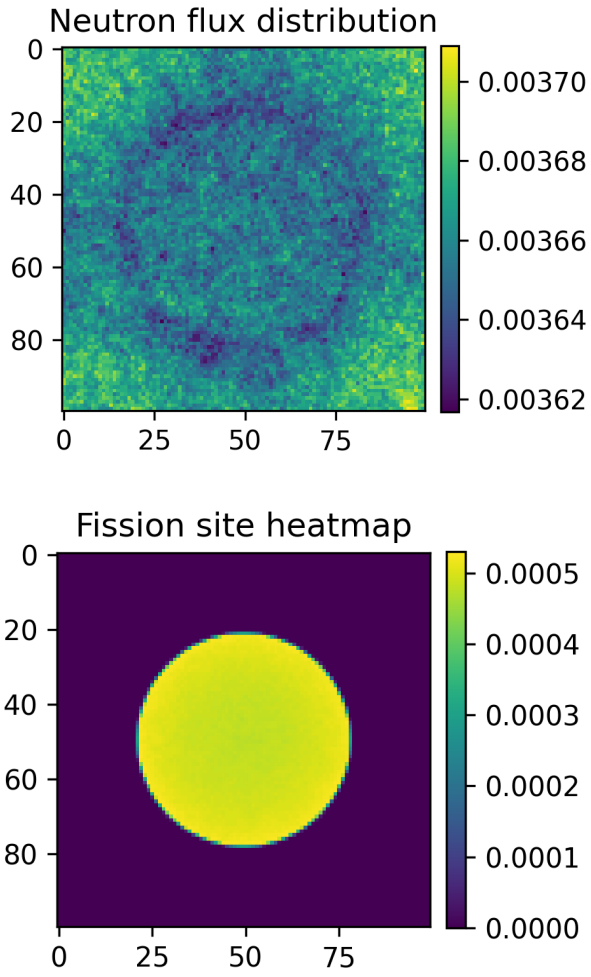


FIG. 4: Above: neutron flux distribution; Below: prompt fission site density. Light-water moderator, reflective boundaries.

The fission neutron spectrum describes neutron energies *at birth*. The spectrum in Fig. 5 exhibits a thermal peak near ∼0.025 eV (the thermal energy reference [7]), resonance depressions in the eV–keV range, and a fast tail at the end. Listing 5 shows how to make the energy filter and how to set up the tally.

Listing 5: Energy-filtered flux tally inside the fuel.

```
# 1e-5 eV ... 10 MeV (200 edges)
energy_bins   = np.logspace(-5, 7, 200)
energy_filter = openmc.EnergyFilter(
    energy_bins)

tally_E = openmc.Tally(name="Energy")
tally_E.filters = [energy_filter]
tally_E.scores  = ["flux"]
```

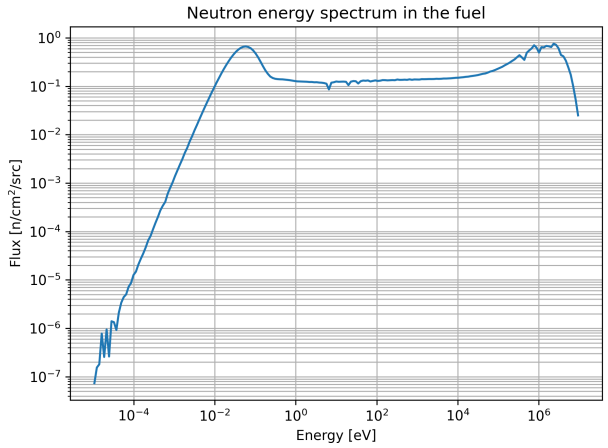


FIG. 5: Fuel neutron flux spectrum with H_2O: track-length flux per source particle vs. energy (log–log).

Moderator/Geometry Study

We replaced the light-water moderator (H_2O) with heavy water (D_2O) while keeping the pin-cell geometry, fuel (UO_2, 2% ^{235}U), and reflective boundaries unchanged. Reflective boundaries suppress leakage entirely, so differences in k_{eff} here stem from effects in the moderator. Table III summarizes the k_{eff} estimators obtained with D_2O; all runs report zero leakage as expected for reflective boundaries. For reference, the earlier light-water case under reflective boundaries yielded $k_{\text{eff}} \approx 1.2915 \pm 0.0009$ (see Table II).

TABLE III: k_{eff} with heavy water (D_2O) moderator; reflective boundaries.

Estimator	k_{eff}	Uncertainty
Combined	**0.870**	± 0.001
Leakage fraction	0.000	± 0.000

Fig. 6 shows the mesh tallies (normalized per source particle): the prompt fission density is confined to the fuel region, whereas the scalar flux extends across materials and exhibits a central depression in the fuel consistent with harder spectra under D_2O. Fig. 7 plots the energy spectrum using an `EnergyFilter`; compared with light water, the spectrum is harder (smaller thermal peak, broader epithermal/fast tail).

Relative to light water (H_2O), heavy water (D_2O) offers much lower absorption but also weaker per-collision [7]. In this small pin-cell with 2% enrichment and reflective boundaries, the limiting factor is *thermalization*: D_2O leaves the spectrum harder inside the fuel, so the ^{235}U fission rate (much larger at thermal energies) is reduced and k_{eff} drops from $\sim$1.29 (H_2O) to $\sim$0.87 (D_2O). The spatial maps and

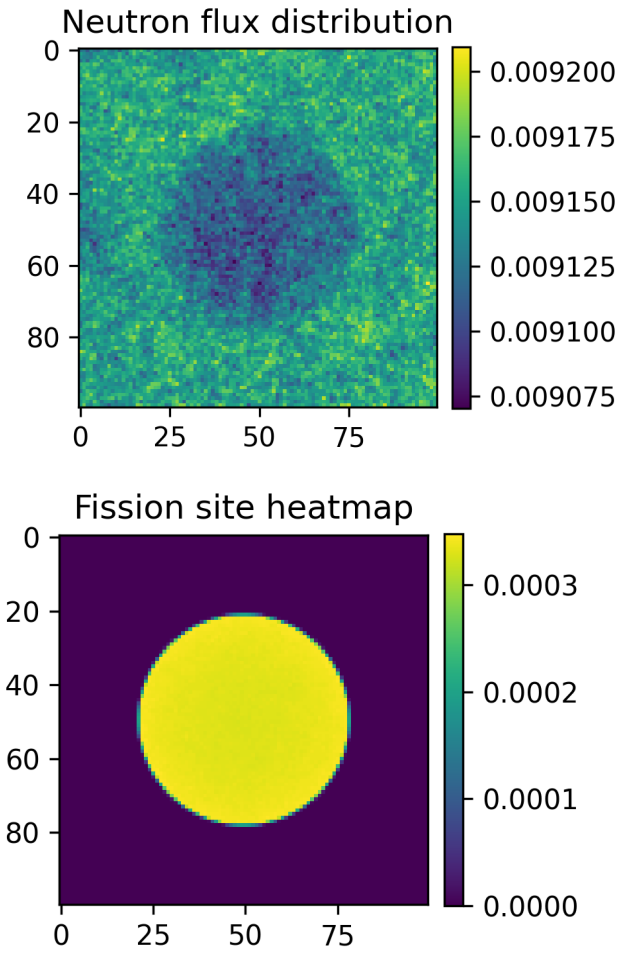


FIG. 6: Above: neutron flux distribution (per source particle). Below: prompt fission site density. Heavy-water moderator (D_2O), reflective boundaries.

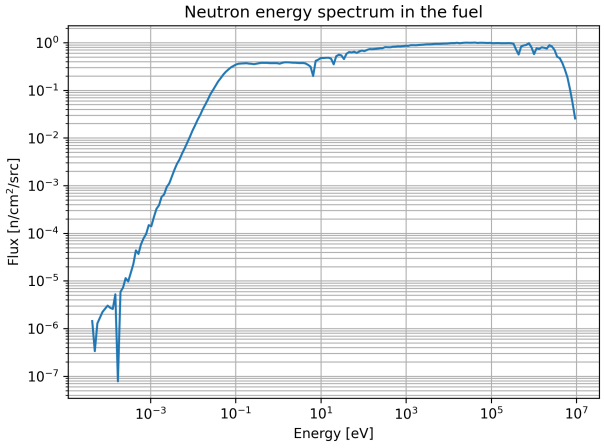


FIG. 7: Fuel neutron flux spectrum with D_2O: track-length flux per source particle vs. energy (log–log).

fuel spectrum are consistent with this picture.

Now, starting from the heavy-water case of Table III, we modified the pin radii as follows, shrinking the fuel (from radii 4.0 mm to 3.8 mm) while increasing both the He gap (from 4.2 mm to 4.4 mm) and the Zr cladding (from 4.5 mm to 4.7 mm):

The resulting k–eigenvalue estimates are shown in Table IV:

TABLE IV: k_{eff} with D_2O and modified radii (smaller fuel, thicker gap/cladding).

Estimator	k_{eff}	Uncertainty
Combined	**0.885**	± 0.001
Leakage fraction	0.000	± 0.000

Relative to the baseline heavy-water pin (Combined $k_{\text{eff}} \approx 0.870$ in Table III), the modified radii increase k_{eff} by $\Delta k \simeq 1.5 \times 10^{-2}$. The effect of reducing the fuel radius is that a larger fraction of slowing-down neutrons avoids capture in the U-238 resonance region before reaching thermal energies [8]. This fuel region reduction gives a positive reactivity contribution.

Thickening the Zr cladding adds some parasitic absorption, while the slightly reduced moderator outside the cladding decreases moderator absorption. With D_2O's intrinsically low absorption, the penalty from Zr is modest, and the net absorption change is small. On the other hand, increasing the He gap (nearly transparent) and cladding thickness slightly delays contact with the moderator, which can harden the spectrum near the fuel. However, the reduced resonance absorption more than compensates for the small uptick in non-fuel captures.

Finally, we returned to the original pin geometry (radii 0.40, 0.42, 0.45 cm) and replaced the D_2O moderator with graphite. Graphite has a lower absorption cross section than light water but a weaker moderating power per collision [8]. With the small pin-cell geometry and 2% enriched UO_2, the resulting k values under reflective boundaries are shown in Table V.

TABLE V: k_{eff} with graphite moderator (original pin geometry; reflective boundaries).

Estimator	k_{eff}	Uncertainty
Combined	**0.5256**	± 0.0006
Leakage fraction	0.000	± 0.000

The combined k_{eff} of ~ 0.526 is significantly below unity and even lower than the D_2O case. This result is physically reasonable: in this tight lattice and with only 2% enrichment, graphite moderates too slowly to provide enough thermal neutrons within the small unit cell. Although graphite's absorption is very low, its lower lethargy gain per collision ($\xi \approx 0.158$ per carbon nucleus [8]) makes it less effective in slowing down neutrons over short distances, so the spectrum remains harder than in H_2O or even D_2O. In larger cores with thicker moderator regions, graphite can indeed sustain criticality, but in this pin-cell calculation it underperforms relative to both light and heavy water.

The three moderators studied (H_2O, D_2O, graphite) illustrate the trade-off between moderating power and absorption. Light water provides the strongest average lethargy gain ($\xi \approx 0.92$ [8]), leading to a very soft spectrum, but hydrogen absorption is relatively high and the moderation ratio is modest ($M \approx 62$ [8]). Heavy water moderates less efficiently ($\xi \approx 0.51$ [8]) but absorbs extremely little, yielding a very large moderation ratio ($M \approx 5830$ [8]). Graphite lies in between ($M \approx 232$ [8]): it has low absorption but weaker slowing-down per unit path length, requiring larger moderator volumes and core sizes to achieve criticality. In our pin-cell results, this physics is reflected by the progression of k: H_2O ~ 1.29, $D_2O \sim 0.87$, and graphite ~ 0.53 under reflective boundaries.

Tuning to Criticality & Four-Factor Links

Starting from the graphite–moderated case, we increased the fuel enrichment in UO_2 to 7% ^{235}U and 93% ^{238}U, expanded the moderator region by enlarging the box from $1.4 \times 1.4 \times 2$ cm^3 to $2 \times 2 \times 2$ cm^3, and slightly reduced the non-fuel radii to $r_{\text{fuel}} = 0.40$ cm, $r_{\text{gap}} = 0.42$ cm, $r_{\text{clad}} = 0.43$ cm (see Fig. 2 for the baseline geometry). Increasing the moderator to fuel volume ratio is beneficial for graphite: carbon absorbs weakly but moderates more slowly than light water; providing more graphite gives additional slowing–down length and softens the spectrum toward thermal energies.

Under reflective boundaries (zero reported leakage), the new k–eigenvalue (near criticality) estimates are shown in Table VI

TABLE VI: k_{eff} with graphite, 7% enrichment, and larger moderator box (reflective boundaries).

Estimator	k_{eff}	Uncertainty
Combined	**0.975**	± 0.001
Leakage fraction	0.000	± 0.000

Because leakage is suppressed by the reflective boundaries, the trend in k_{eff} here can be interpreted in terms of the infinite-lattice multiplication $k_\infty = \eta\, f\, p\, \varepsilon$ (the *four–factor formula*) [2]. The specific changes we made act on these factors as follows:

- Reproduction factor η: Raising the ^{235}U fraction increases the number of fission neutrons produced per absorption in the fuel, directly boosting η [2].
- Resonance–escape probability p: Increasing the moderator volume provides additional slowing–down distance in graphite, reducing the chance of resonance captures in ^{238}U and hence increasing p [2].
- Thermal utilization f: Two competing effects act on f; enlarging the moderator increases total thermal absorption outside the fuel, which by itself would *lower* $f = \langle\Sigma_a\phi\rangle_{\text{fuel}}/\langle\Sigma_a\phi\rangle_{\text{total}}$. However, the higher enrichment raises the fuel absorption relative to the moderator and reduces parasitic captures in the cladding region after we thinned it, partially offsetting the above and helping f [2].
- Fast–fission factor ε: The geometry changes applied cause comparatively small variation in ε for a thermal lattice; the dominant gains here come from η and p [2].

Overall, enrichment ($\uparrow$ η) together with a larger graphite moderator volume ($\uparrow$ p) moved the graphite case from a markedly subcritical lattice to near–critical ($k_{\text{eff}} \approx 0.975$). This behavior is consistent with the qualitative roles of the four factors in thermal systems [2].

The top-view of the new pin cell with graphite moderator is shown in Figure 8. Note the differences between Figure 2 and Figure 8.

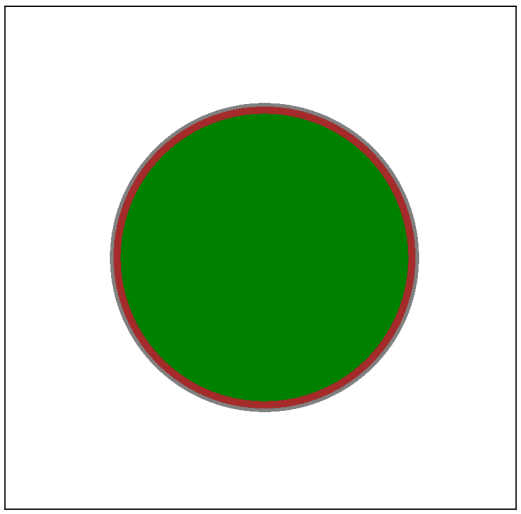


FIG. 8: Top view of the pin-cell (xy): fueld (green), Helium gap (brown), cladding (grey) and moderator (white).

Thermal Utilization Factor f vs. Pin Size

In an infinite (no-leakage) thermal lattice, the thermal utilization factor is

$$f \equiv \frac{\langle\Sigma_a\phi\rangle_T^F}{\langle\Sigma_a\phi\rangle_T},$$

that is, the fraction of *thermal* neutron absorptions that occur in the fuel (F) relative to all thermal absorptions in the system (T) [1]. Here we take the thermal range to be $E \in [0, 1]$ eV, following the common "thermal group" convention used in reactor analysis [8]. Reflective boundaries are used so there is no leakage.

We compute f directly from reaction–rate tallies: (1) a numerator tally of thermal absorption restricted to the fuel cell, and (2) a denominator tally of thermal absorption over the whole model (see Listing 6). To study "pin size", we vary the *lattice pitch*—the width/height of the rectangular unit cell that contains fuel, gap, cladding, and moderator—while keeping the pin radii fixed. This varies the moderator–to–fuel volume ratio without altering the pin itself. For each pitch we run a fresh eigenvalue calculation, read the statepoint, compute f, and store (V, f) with $V = \text{pitch}^2 \times \text{height}$. For these f calculations, we impose the original materials and enrichment (UO_2, 2% ^{235}U fuel).

Listing 6: Thermal absorption tallies for $f = \langle\Sigma_a\phi\rangle_T^F/\langle\Sigma_a\phi\rangle_T$ with a thermal window $E \in [0, 1]$ eV.

```
# Thermal window [0, 1 eV]
thermal_E = openmc.EnergyFilter([0.0, 1.0])

# Numerator: thermal absorption IN FUEL
t_num=openmc.Tally(name="therm_abs_in_fuel")
t_num.filters = [fuel_only, thermal_E]
t_num.scores  = ["absorption"]

# Denominator: general thermal absorption
t_den = openmc.Tally(name="therm_abs_total")
t_den.filters = [thermal_E]
t_den.scores  = ["absorption"]
```

Figures 9 and 10 show the measured trend of f with pin-cell volume using H_2O and D_2O as moderators, respectively.

For a fixed fuel pin geometry (UO_2 radius 0.4 cm, He gap 0.02 cm, Zircaloy cladding up to 0.45 cm), increasing the surrounding water volume while keeping these radii constant leads to a decrease in the thermal utilization factor f. In the light water case, the larger moderator region enhances thermalization, producing a higher thermal flux in the water, but also increases the probability of thermal neutron absorption by hydrogen. As a result, a smaller fraction of thermal absorptions occur in the fuel as we increase the volume.

Conversely, if light water (H_2O) is replaced by heavy water (D_2O), f rises significantly (in comparison with H_2O) because deuterium has a much lower absorption cross section; thus, the moderator contributes less to the total absorption balance. However, as the moderation volume increases while maintaining the fuel volume, f will still decrease.

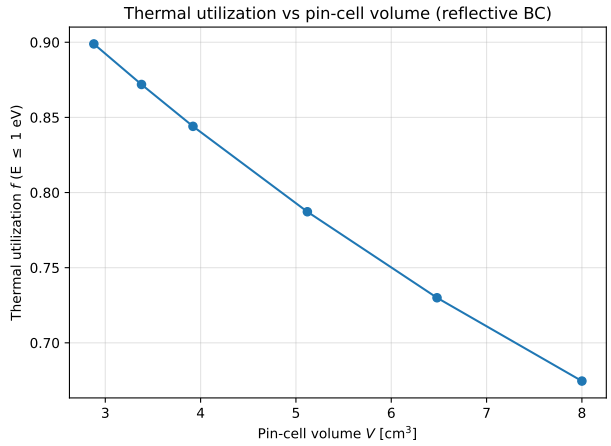


FIG. 9: Thermal utilization f versus pin-cell volume V (reflective boundaries and H_2O moderator). Thermal band $E \in [0, 1]$ eV used in the tallies.

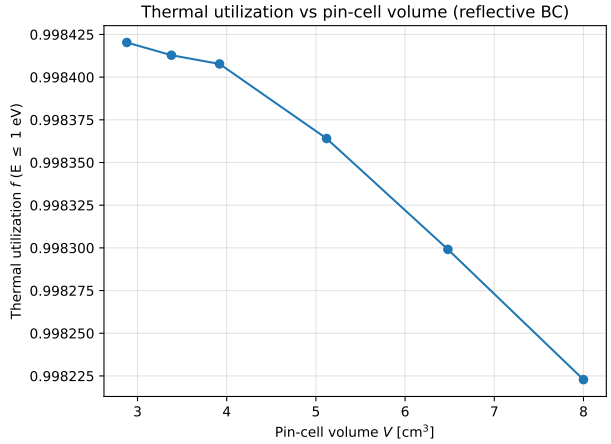


FIG. 10: Thermal utilization f versus pin-cell volume V (reflective boundaries and D_2O moderator). Thermal band $E \in [0, 1]$ eV used in the tallies.

From pin-cell to a small core

17×17 Fuel Assembly with Control-Rod Positions

In this section, the single pin-cell model developed earlier was extended to build a small reactor core, represented by a 17×17 fuel assembly in the xy-plane. Each pin cell, defined previously, consists of concentric regions representing the fuel pellet (UO_2), helium gap, zirconium alloy cladding, and the surrounding light-water moderator. These individual pin-cells were arranged in a rectangular lattice using the `openmc.RectLattice` class, as shown in Fig. 11. Reflective boundary conditions were applied on all outer surfaces, effectively simulating an infinite lattice of assemblies.

Nine central pin-cells were replaced with control rods made of boron carbide (B_4C). The control-rod material was chosen because boron, particularly ^{10}B, exhibits a very high neutron absorption cross section in the thermal energy range. The chosen B_4C material was modeled with a density of 2.52 g/cm^3 [6] using 19.9% ^{10}B and 80.1% ^{11}B. The high neutron-capture probability of boron makes B_4C an effective absorber.

Fig. 12 shows the macroscopic cross sections of B_4C as a function of neutron energy. The dominant absorption reactions are (n, α) (MT 107) and (n, γ) (MT 102). At thermal energies, the (n, α) reaction dominates, confirming that B_4C is well-suited for control-rod applications in thermal reactors such as pressurized water reactors (PWRs).

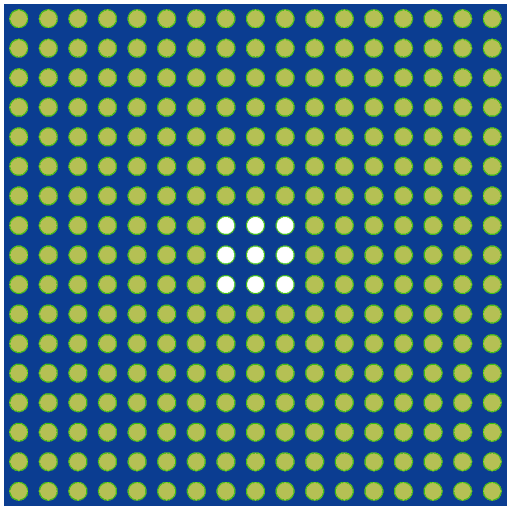


FIG. 11: Top view of the 17×17 fuel assembly with nine central control rods made of B_4C (white). The background (blue) represents the light-water moderator, and the green circles correspond to UO_2 fuel pins.

Core Construction & Critical Tuning

Starting from the validated pin-cell model (UO_2 fuel, He gap, Zircaloy cladding, H_2O moderator), we assembled a small pressurized-water-reactor (PWR)

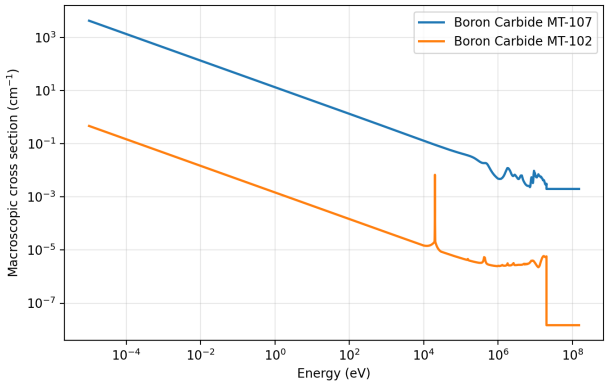


FIG. 12: Macroscopic neutron absorption cross sections of boron carbide (B_4C) for the (n,α) (MT 107) and (n,γ) (MT 102) reactions. [9]

core by tiling complete 17×17 fuel-assembly universes on a rectangular lattice and then trimming the lattice to a circular core region. The core (105 cm) is radially surrounded by a light-water reflector (25 cm) and an outer steel vessel (10 cm); the exterior boundary is set to `vacuum`. Five assemblies include a central 3×3 cluster of B_4C control-rod pins (yellow squares in Fig. 13), arranged in a "cross" pattern. All other assemblies are fuel-only. The pin-cell compositions and dimensions match those defined in the earlier sections of this report.

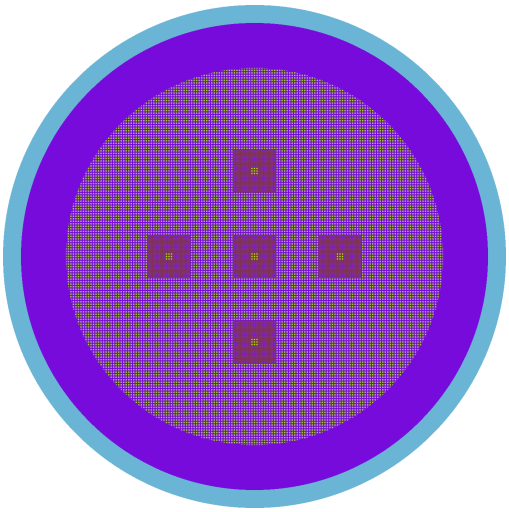


FIG. 13: Top view (XY, $z = 0$) of the modeled core: fuel assemblies (purple), water reflector (violet ring), steel vessel (cyan), and the five assemblies with 3×3 B_4C control-rod clusters (yellow centers).

Assemblies are first packaged as `Universes` composed of a 3-D `RectLattice` of pin universes (fuel or B_4C). A second `RectLattice` places these assembly universes into an $N_x \times N_y$ grid; a circular region (`ZCylinder`) carves the core boundary, with concentric regions for the water reflector and the steel vessel.

With the above configuration (fuel-only assemblies except for the five 3×3 B_4C clusters, water reflector, steel vessel, vacuum outer boundary and 2% ^{235}U), the measured k_{eff} estimators and leakage fraction are shown in Table VII.

TABLE VII: k_{eff} for the core with light water moderator and Fig. 13 structure.

Estimator	k_{eff}	Uncertainty
Combined	**1.2532**	± 0.0004
Leakage fraction	0.00420	± 0.00005

The core is supercritical ($\Delta k \approx +0.253$) with small but nonzero leakage, consistent with a finite reflector and a vacuum vessel boundary. The water reflector reduces radial losses ("reflector savings"), while the limited number and axial extent of control pins provide only a modest negative reactivity.

To bring the system to a critical state (lower k_{eff}), we consider increasing the number of B_4C control-rod pins as shown in Fig. 14. The measured k_{eff} estimators and leakage fraction are shown in Table VIII.

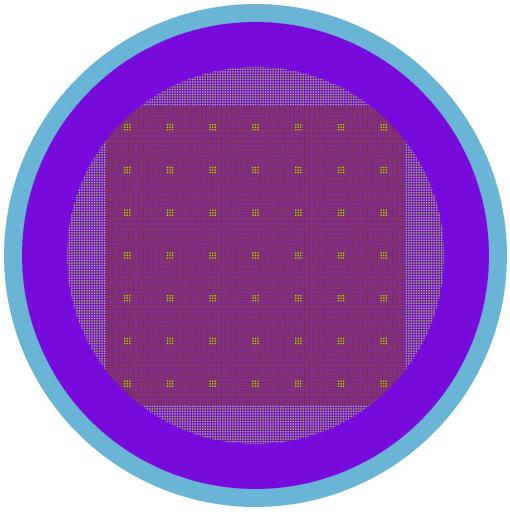


FIG. 14: Top view (XY, $z = 0$) of the modeled core: fuel assemblies (purple), water reflector (violet ring), steel vessel (cyan), and the forty nine assemblies with 3×3 B_4C control-rod clusters (yellow centers).

TABLE VIII: k_{eff} for the core with light water moderator and Fig. 14 structure.

Estimator	k_{eff}	Uncertainty
Combined	**1.1702**	± 0.0005
Leakage fraction	0.00414	± 0.00006

Fuel Temperature effect with Doppler Broadening

Using the small-core geometry shown in Fig. 14, we performed a fuel-temperature sweep while keeping all other model inputs unchanged. The fuel temperature T_f was varied from 400 K to 1200 K in discrete steps, and for each case an eigenvalue calculation was run to obtain k_{eff} and its uncertainty.

This temperature interval covers typical PWR operating and transient conditions for UO_2 fuel: values near 400–600 K represent low-power states, 700–1000 K cover nominal hot conditions, and 1100–1200 K provide [10] margin to explore the temperature-feedback trend into elevated temperatures for Doppler studies.

Fig. 15 shows a clear decrease of k_{eff} with increasing T_f. This is related to the *Doppler broadening* of resonance absorption. As T_f rises, the relative motion of target nuclei widens and the resonance profiles of the cross sections get broader. Although the peak cross section diminishes, the broadened resonances interact with a wider portion of the neutron energy distribution, increasing the probability of resonance absorption in the fuel before neutrons fully thermalise [11]. In the four-factor picture, this manifests mainly as a reduction of the *resonance escape probability* p. Consequently, the reactivity becomes more negative with temperature. This effect provides inherently stabilising feedback in thermal reactors: a rise in power heats the fuel, enhances Doppler absorption, and lowers k_{eff}, counteracting the initial increase.

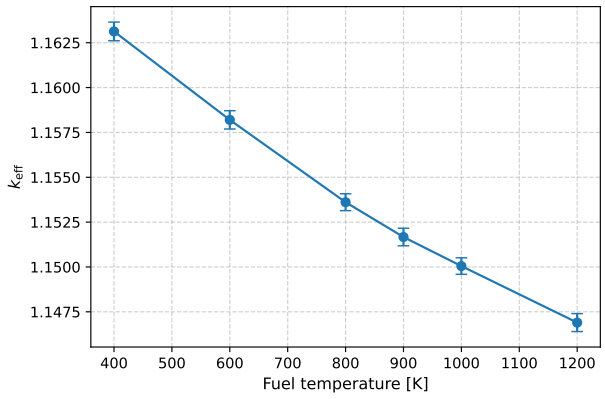


FIG. 15: Calculated k_{eff} as a function of fuel temperature for the core of Fig. 14.

Core Flux Distribution

To obtain the spatial neutron-flux map for the full core (see Fig. 16), we tallied the energy-integrated scalar flux on a 2D regular mesh covering the circular core, reflector, and vessel. The mesh (XY, z-integrated) was applied via an `openmc.RegularMesh` + `MeshFilter` and a single `flux` score; tallies were exported to XML, the k-eigenvalue simulation was run, and the statepoint was post-processed to reshape the tally to (n_x, n_y) and plot it as a contour image.

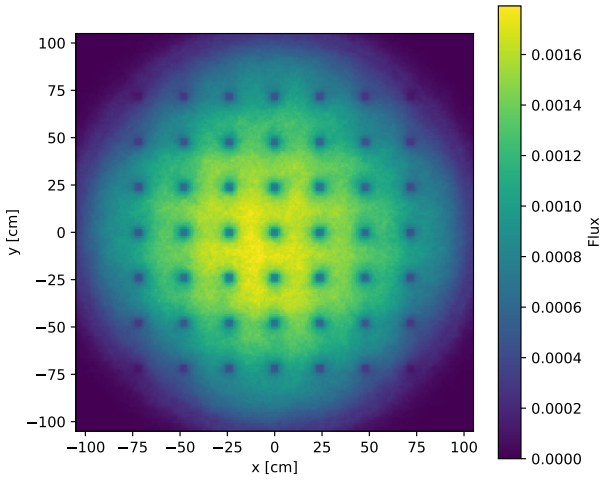


FIG. 16: Energy-integrated neutron flux (XY, z-integrated) over the modeled core.

The map exhibits the expected radial shape: higher flux in the inner core and reduced levels near the periphery due to leakage and the presence of reflector/vessel. Local depressions arise where control rods are present.

Six-Month Depletion at 1200 MW(th)

For the burnup study, minor modifications were introduced to the reactor configuration previously described. The enrichment of the ^{235}U in the UO_2 fuel was reduced from 2.0% to 1.2%, and all B_4C control rods were removed from the core. This fuel change was necessary to bring the system near criticality once the control materials were withdrawn. With these adjustments, the reactor was allowed to operate at a fixed thermal power of 1200 MW(th) for a simulated period of six months using the OpenMC depletion interface.

Fig. 17 shows the calculated evolution of the effective multiplication factor k_{eff} over time. The reactor initially operates in a slightly supercritical condition with $k \approx 1.08$, reaching criticality ($k = 1.00$) after about ninety days of operation. As fuel burnup progresses and fissile material is consumed, the

system becomes progressively less reactive. After six months, the reactor becomes subcritical with $k \approx 0.95$, as expected from the combined effects of ^{235}U depletion, plutonium buildup, and accumulation of fission products.

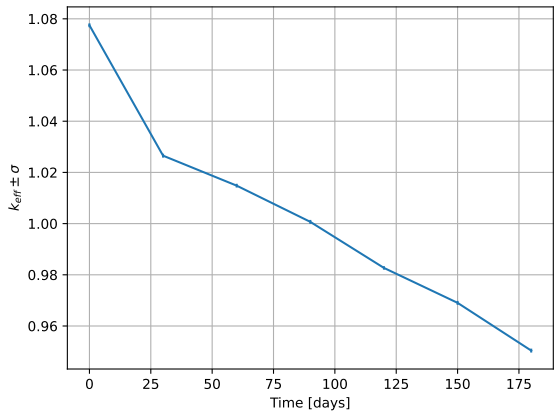


FIG. 17: Evolution of k_{eff} during six months of operation at 1200 MW(th).

Conversion Ratio $^{239}Pu/^{235}U$

The conversion ratio (CR) is defined here as the ratio of newly formed ^{239}Pu atoms ($N_{^{239}\text{Pu}}$) to the number of ^{235}U atoms ($N_{^{235}\text{U}}$) consumed over the depletion interval:

$$\text{CR} = \frac{\Delta N_{^{239}\text{Pu}}}{-\Delta N_{^{235}\text{U}}} = \frac{N_{^{239}\text{Pu}}^{\text{final}} - N_{^{239}\text{Pu}}^{\text{initial}}}{N_{^{235}\text{U}}^{\text{initial}} - N_{^{235}\text{U}}^{\text{final}}}.$$

From the results (Fig. 18), the initial/final inventories are:

$$N_{^{235}\text{U}}^{\text{initial}} = 1.02 \times 10^{27}\,,\; N_{^{235}\text{U}}^{\text{final}} = 5.09 \times 10^{26}$$

$$N_{^{239}\text{Pu}}^{\text{initial}} = 0\,,\; N_{^{239}\text{Pu}}^{\text{final}} = 2.25 \times 10^{26}.$$

Hence,

$$-\Delta N_{^{235}\text{U}} = 5.11 \times 10^{26}, \qquad \Delta N_{^{239}\text{Pu}} = 2.25 \times 10^{26}$$

$$\text{CR} \approx \frac{2.25}{5.11} \simeq 0.44.$$

A conversion ratio of ~ 0.44 indicates that the thermal core converts a substantial fraction of ^{238}U into ^{239}Pu during the first six months, yet it remains far from breeding (CR < 1). Physically, this Pu build-up partially offsets the loss of ^{235}U reactivity later in the cycle, but the net effect over the period is a decrease in k_{eff} as observed. The comparable orders of magnitude of the end-of-period inventories ($N_{^{239}\text{Pu}}$ vs. $N_{^{235}\text{U}}$) highlight that plutonium contributes meaningfully to fission towards the end of the interval.

With a conversion ratio of roughly 0.44, the core generates a non-negligible inventory of ^{239} Pu during the six-month campaign. Even though this plutonium is of reactor-grade composition and is not well suited to weapons applications, its presence is nonetheless proliferation-sensitive and mandates rigorous accounting, secure storage and regulatory oversight.

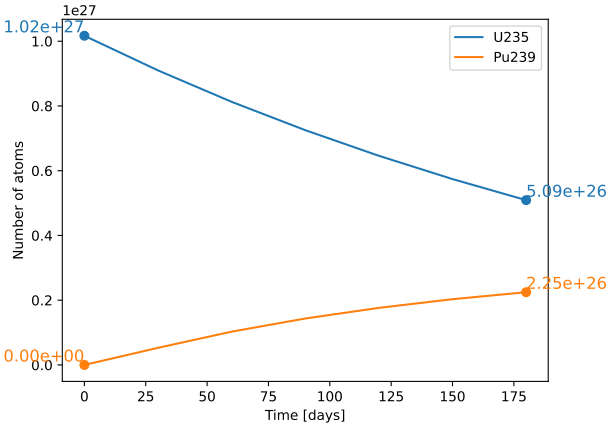


FIG. 18: Total number of atoms of ^{235}U and ^{239}Pu vs. time during the six-month depletion.

Neutron Density Estimate

We can estimate the steady neutron density by relating prompt fission power to the total number of neutrons in the core. Assuming that 95% of the thermal power comes from the prompt energy of new fissions, each fission releases $E_p = 190\,\text{MeV} = 3.044 \times 10^{-11}\,\text{J}$, the average neutron yield is $\nu \simeq 2.4$, and the average neutron lifetime is $\ell = 50\,\text{ms}$, the prompt power satisfies

$$P_{\text{prompt}} = \frac{E_p}{\nu\,\ell}\,N \quad \Rightarrow \quad N = \frac{0.95\,P_0\,\nu\,\ell}{E_p}.$$

Dividing by the active volume (fuel volume), $V 3.463\,492\,803\,247\,218 \times 10^6\,\text{cm}^3$, gives the neutron density $n = N/V$.

With $P_0 = 1200\,\text{MW(th)} = 1.2 \times 10^9\,\text{W}$, $\nu = 2.4$, $\ell = 0.050\,\text{s}$, and $E_p = 3.044 \times 10^{-11}\,\text{J}$,

$$n \;\approx\; 1.3 \times 10^{12}\,\frac{N}{\text{cm}^3}.$$

Using the *fuel* volume yields a higher (conservative) density because it excludes moderator and structure. If instead one used the full active *core* volume (fuel + moderator within the core boundary), the estimate would decrease. In either case, the result is of the correct order of magnitude for a 1200 MW(th) thermal core under the exercise assumptions.

Safety Features

The model exhibits strong negative Doppler feedback in UO_2: increasing fuel temperature broadens resonances, reducing the resonance-escape probability p and driving k_{eff} down with temperature (see Fig. 15). This stabilising mechanism is evident in the fuel-temperature sweep and is a cornerstone of inherent safety in thermal reactors.

In the assembly/core models we reserved positions for B_4C control rods, which should provide rapid shutdown. However, in the simulations presented, the control rods don't have a rapid reactivity reduction as they are composed of natural boron (19.9% ^{10}B and 80.1% ^{11}B). Increasing the ^{10}B enrichment over ^{11}B would be helpful in an emergency shutdown case. Adding soluble boron to the moderator would also be helpful, as it would supply an additional, spatially uniform negative reactivity to keep the core critical over burnup while preserving shutdown margin.

The following features would require verification: (i) shutdown margin with the rod stuck and measured rod-drop times; (ii) power-shape/peaking factors and thermal limits under representative operating states; (iii) xenon/samarium dynamics and stability; (iv) mechanical/materials aspects (cladding temperatures) under limiting events.

CONCLUSION

We developed a reproducible OpenMC workflow that progresses from a validated pin-cell to a small, reflected PWR-type core and used it to interrogate key neutronic behaviours. The assembled core with light-water reflector is supercritical without controls, whereas introducing distributed control-rod clusters reduces reactivity as expected. Fuel-temperature sweeps show a clear, negative Doppler coefficient consistent with broader resonance absorption at higher temperatures. Full-core flux maps display the anticipated radial shape and local depressions under control-rod locations. A six-month depletion at 1200 MW(th)—with reduced ^{235}U enrichment and rods withdrawn—shows k_{eff} declining from slightly supercritical to ~ 0.95, matching the combined effects of fissile depletion, Pu build-up, and fission-product poisoning. From inventories we obtain CR ≈ 0.44 for ^{239}Pu/^{235}U, i.e., significant conversion but no breeding, and an order-of-magnitude neutron density near 10^{12} n cm^{-3}. Overall, the results are self-consistent across methods (eigenvalue, tallies, depletion) and align with thermal-reactor intuition.

[1] University of Oslo Department of Physics. Project description: Introduction to nuclear reactor physics, 2025.

[2] James J. Duderstadt and Louis J. Hamilton. *Nuclear Reactor Analysis*. John Wiley & Sons, 1976.

[3] Jon Shimwell. Nuclide microscopic cross section plotter. https://openmc-data-storage.github.io/nuclide_cross_section_plotter.rs/index.html#/, 2025. Accessed: 2025-10-26.

[4] The OpenMC Team. Openmc beginner's guide. https://docs.openmc.org/en/stable/usersguide/beginners.html, 2025. Accessed: 2025-09-22.

[5] PubChem. Pubchem substance (sid 329798917), 2025. Accessed 22 Sep 2025.

[6] PubChem. Periodic table: Density, 2025. Accessed 22 Sep 2025.

[7] John R. Lamarsh and Anthony J. Baratta. *Introduction to Nuclear Engineering*. Prentice Hall, 2001.

[8] S. Okajima, T. Kugo, and T. Mori. *Nuclear Reactor Physics*. Springer, 2024.

[9] Jon Shimwell. Materials macroscopic cross section plotter. https://fusion-neutronics.github.io/materials_for_mc_online/, 2025. Accessed: 2025-10-26.

[10] U.S. Nuclear Regulatory Commission, Technical Training Center. *Pressurized Water Reactor (PWR) Systems*. USNRC Technical Training Center, 2003. Reactor Concepts Manual, Chapter 4.

[11] Serge Marguet. *The Physics of Nuclear Reactors*. Springer, 2017.